

ZaraiLink — Demo

Brief

Working document for Salman Adnan

July 7, 2026

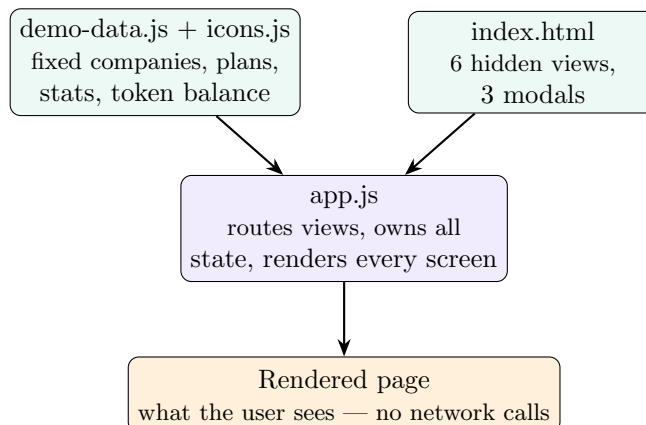
Honesty check

This documents the **demo codebase** (`zarailink-demo`) only: a scripted, fabricated-data product tour with zero live network calls. The private team project it visually reproduces — a three-person final-year build (Django/DRF backend, React frontend, NLU pipeline, hybrid retrieval, learning-to-rank, over real Pakistani customs trade data), led by Salman Adnan (64 of 97 commits) with real, substantial work by teammates Umar Kashif and Fahad Nadeem — is not published here because full-source release needs every author’s consent, which has not yet been given. This demo exists so the work can be shown without that consent problem. All facts below come from the demo’s own files.

1 What it is

A single static HTML/CSS/JS page (no backend, no build step, no framework) that behaves like a small multi-screen app: six views — Dashboard, Search Results, Trade Directory, Watchlist, Company Profile, Subscription — switched between with plain JavaScript inside one already-loaded page, all built from source read directly off the real app’s actual React components (named explicitly in the demo’s own comments). Four fabricated companies with fabricated contacts, products, and trade statistics stand in for real customs data. The centerpiece is a token-gated contact-unlock economy that mirrors the real app’s per-contact (not all-or-nothing) unlock design.

2 Architecture



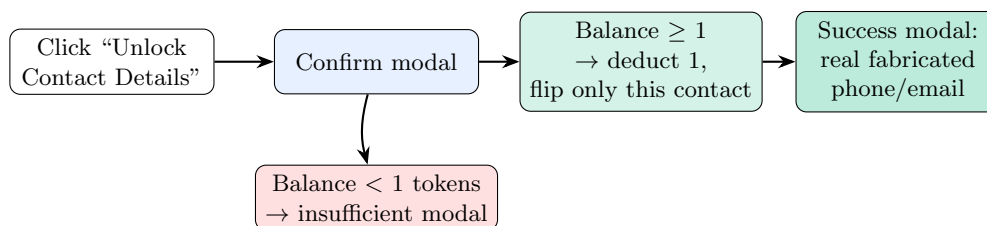
Nine files total. All state — which view is active, which contacts are unlocked, the token balance, the watchlist — lives in plain JavaScript variables in browser memory and is lost on refresh; there is no database, no `localStorage`, no server.

3 The six views

Every screen already exists in `index.html` as a hidden `<section class="view">`; `switchView()` just moves an `is-active` class between them, so there is no page reload.

View	What it reproduces
Dashboard	Search home: “4 Ways to Search” card, live search-mode pill, Import/Export toggle, stats strip, features grid.
Search Results	Sticky header, product-filter sidebar, 4 trade-direction pills, result cards, token-gated category unlock banner.
Trade Directory	Searchable company-card grid with country/sector/type badges and a watchlist star.
Watchlist	Starred companies, reusing the Directory’s card renderer.
Company Profile	Stats, products, and the per-contact masked-data unlock UI.
Subscription	Three plan cards with token allotments.

4 Key flow: unlocking a contact



A locked contact row is both text-masked (e.g. `●●●●●@example.com`) and CSS-blurred; unlocking spends exactly 1 token and flips *only* that one contact — every other locked contact on the same page stays locked, faithfully matching the real app’s per-contact (not per-company) unlock model.

One fabricated contact is marked “public” and always shows real data at no token cost, distinct from a token-unlocked contact. Search Results has a second, category-level unlock (5 tokens for the whole result set) with its own confirm-free single-click flow; its button copy reads “Unlock for 500 Tokens” but the code actually deducts 5 — a real copy/logic mismatch in the demo as written, noted here rather than glossed over.

5 What’s simulated vs. what’s real

- **Real, faithfully reproduced:** the six screens’ actual structure and component layout (read from the real app’s source files), the per-contact unlock mechanics, the masking pattern, the trade-direction pill framing, the search-mode detection rule (digit → HS code; scope selected → AI query; else product search).
- **Simulated / fabricated:** all company, contact, and trade data; the token balance and every deduction target a fixed starting value of 7; “search” always filters the same four companies by simple hard-coded rules, not real NLU, hybrid retrieval, or learning-to-rank.
- **Not represented at all:** the real app’s Trade Intelligence cluster — link prediction, node2vec similar-company matching, and GNN (Graph Neural Network) work — which the demo’s own README calls the product’s most technically distinctive piece and names as the natural next screen to add.

6 Styling

`styles.css` states its own method: colors and per-screen layout are copied 1:1 from the real app’s own `index.css`, read directly from its actual component files, because that is pure visual design language any browser already renders when it loads the real app — not its private backend logic (NLU, ranking, real trade data, or API routes), none of which appear here. The palette is a light SaaS-dashboard theme (emerald primary, white/near-white surfaces) with a dark slate navbar and two dark bands, distinct from the ReelFlow demo’s all-dark console theme. Fonts (a self-hosted Inter variable font) and icons (hand-drawn SVGs matching the real app’s Lucide icon set) are also reproduced to avoid falling back to emoji, which read as a lower-fidelity visual language.

7 Team-authorship honesty

Restated plainly: ZaraiLink is a three-person team’s final-year project. Salman Adnan led it and wrote most of the commits, but two named teammates (Umar Kashif, Fahad Nadeem) contributed real, substantial work. None of their code or any real project artifact is in this demo repository — the demo’s screens, styling, and scripted logic were built from scratch in this separate, public repository by reading the real app’s rendered structure, not by copying its private source.

8 Glossary

Static page

HTML/CSS/JS served as-is, with no server generating content per request.

Event delegation

One click listener on a large container, inspecting which child was actually clicked.

NLU Natural Language Understanding: extracting structured meaning from free text. Used by the real app; simulated here only as simple pattern matching.

HS code Harmonized System code, a standardized numeric customs classification for product categories.

Token economy

Users spend a limited resource to unlock premium actions; here, 1 token per contact.

GNN Graph Neural Network: a machine-learning model architecture for learning from graph-structured data (e.g. a network of trading companies).

Design tokens

Named, reusable CSS values (colors, radii, shadows) defined once and reused throughout a stylesheet.

9 Facts at a glance

Aspect	This demo
Backend	None; entirely client-side.
Data	4 fabricated companies, 3 subscription plans, fixed stats, in <code>demo-data.js</code> .
State	Module-level JS variables, lost on refresh.
Views	6, toggled by CSS class, no reloads.
Unlock costs	1 token/contact; 5 tokens/category (button text says 500 — a copy bug).
Hosting	Static publish via Netlify.
License	PolyForm Strict 1.0.0.
Why it exists	Team project; full-source publication awaits all authors' consent.